

1. **Experiment:** Lentiviral Transduction of A2B3C3-FcR γ in Raji Cells and Puromycin Selection
2. **Time:** 2026.02.21-2026.03.06
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4. **Materials:** Hek293T cells, Raji cells, Hieff Trans® Polyethylenimine Linear (PEI), pCDH-A2B3C3-FcR γ transfer plasmid (A2B3C3-FcR γ fused as a single complete target protein), pMD2.G envelope plasmid, psPAX2 packaging plasmid, DMEM high glucose, RPMI-1640 medium, 10% heat-inactivated FBS, 1% penicillin-streptomycin, Polybrene, and Puromycin

5. **Method:**

- (1) One day before transfection, Hek293T cells were passaged and evenly plated on 100 mm culture dishes (at 6×10^6 cells per dish). Cells were cultured at 37 °C in a 5% CO₂ incubator. The medium was replaced with 10.5 mL of pre-warmed, antibiotic-free medium containing 5% FBS for 3 h before transfection.
- (2) A mixture of three plasmids (transfer plasmid + envelope plasmid + packaging plasmid) in equimolar amounts was prepared, and transfection was performed using the PEI transfection method with the Hieff Trans® Polyethylenimine Linear. The transfection mixture composition per well was as follows:

Table 1. Transfection mixture composition.

Transfection tube	Reagent	Volume/Amount
A	PEI (1 mg/mL)	90 μ L
	Antibiotic-free medium	910 μ L (to 1 mL total)
	Transfer plasmid (pCDH)	15 μ g
B	Envelope plasmid (pMD2.G)	5 μ g
	Packaging plasmid (psPAX2)	10 μ g
	Antibiotic-free medium	Up to 1 mL

- (3) Puromycin kill curve determination: Raji cells were seeded in 24-well plates at 2×10^5 cells/mL in 1 mL RPMI complete medium per well. Puromycin was added at graded concentrations of 0, 0.25, 0.5, 0.75, 1.0, 1.5, 2.0, 3.0, 4.0, and 5.0 μ g/mL. Cells were incubated at 37 °C, 5% CO₂ for 72 h, and viable cells were counted at 24 h, 48 h, and 72 h.
- (4) The PEI solution (Tube A) and the plasmid mixture (Tube B) were gently mixed separately, then combined by adding Tube B dropwise into Tube A using a micropipette. The mixture was vortexed briefly and incubated at room temperature for 15 min to allow complex formation. The transfection complexes were then added dropwise to the Hek293T cells, and the plates were returned to the incubator.
- (5) Viral supernatant collection: At 48 h and 72 h post-transfection, collect the viral supernatant from Hek293T cells, filter through a 0.45 μ m filter, and use immediately or store at -80 °C.
- (6) Cell Inoculation:

① Collect Raji cells from a culture flask.

② Centrifuge at $300 \times g$ for 5 min at room temperature.

③ Carefully remove the supernatant and resuspend the cell pellet in an appropriate volume of complete medium (RPMI-1640 with 10% FBS and 1% penicillin-streptomycin).

④ The cell density was determined using an automated cell counter, and the suspension was diluted to a final concentration of 1×10^6 cells/mL.

⑤ Add the calculated volume of cells to an appropriate volume of complete medium to achieve a final concentration of 1×10^6 cells/mL.

⑥ For each well of a 6-well plate (or equivalent dish), 2 mL of this cell suspension (2×10^6 cells) was dispensed. If using a 10 cm dish, adjust the volume to 10 mL (1×10^7 cells). The plates were incubated at 37 °C in a 5% CO₂ incubator for 24 h prior to infection.
- (7) Infection: To each well, Polybrene was added to a final concentration of 8 μ g/mL. The multiplicity of

infection (MOI) was set to 0.3, using the volume calculated as $(0.3 \times \text{cell number}) / \text{virus titer}$, was added to the wells. The plates were then incubated at 37 °C for 12 h, then replace the viral inoculum with fresh complete medium.

- (8) Puromycin selection: At 60 h post-transduction, cells were collected by centrifugation at $300 \times g$ for 5 min and discard the culture supernatant. The cell pellet was resuspended in fresh RPMI complete medium containing Puromycin at the working concentration determined in step (3). Cells were cultured at 37 °C, 5% CO₂, and the medium was replaced with fresh Puromycin-containing medium every 2 days. Selection was continued for 10 days until all non-transduced control cells (without virus, with Puromycin) were dead.
- (9) Stable cell line expansion and validation: Surviving cells were expanded in RPMI complete medium containing maintenance Puromycin (0.75 µg/mL) for 4 passages. Puromycin-resistant cells were stained with fluorescent anti-Flag or anti-human Fc antibody, Fc-positive cells were enriched by flow cytometric sorting, and the sorted cells were expanded for downstream experiments.

6. Result:

After the first round of screening identified A2B3C3 as the dominant combination, we introduced A2B3C3-Fc into Raji cells via lentiviral transduction and obtained stably transduced Raji cell pools through antibiotic selection. This cell pool can be used for subsequent expression validation of A2B3C3-Fc and the second round of evolution.